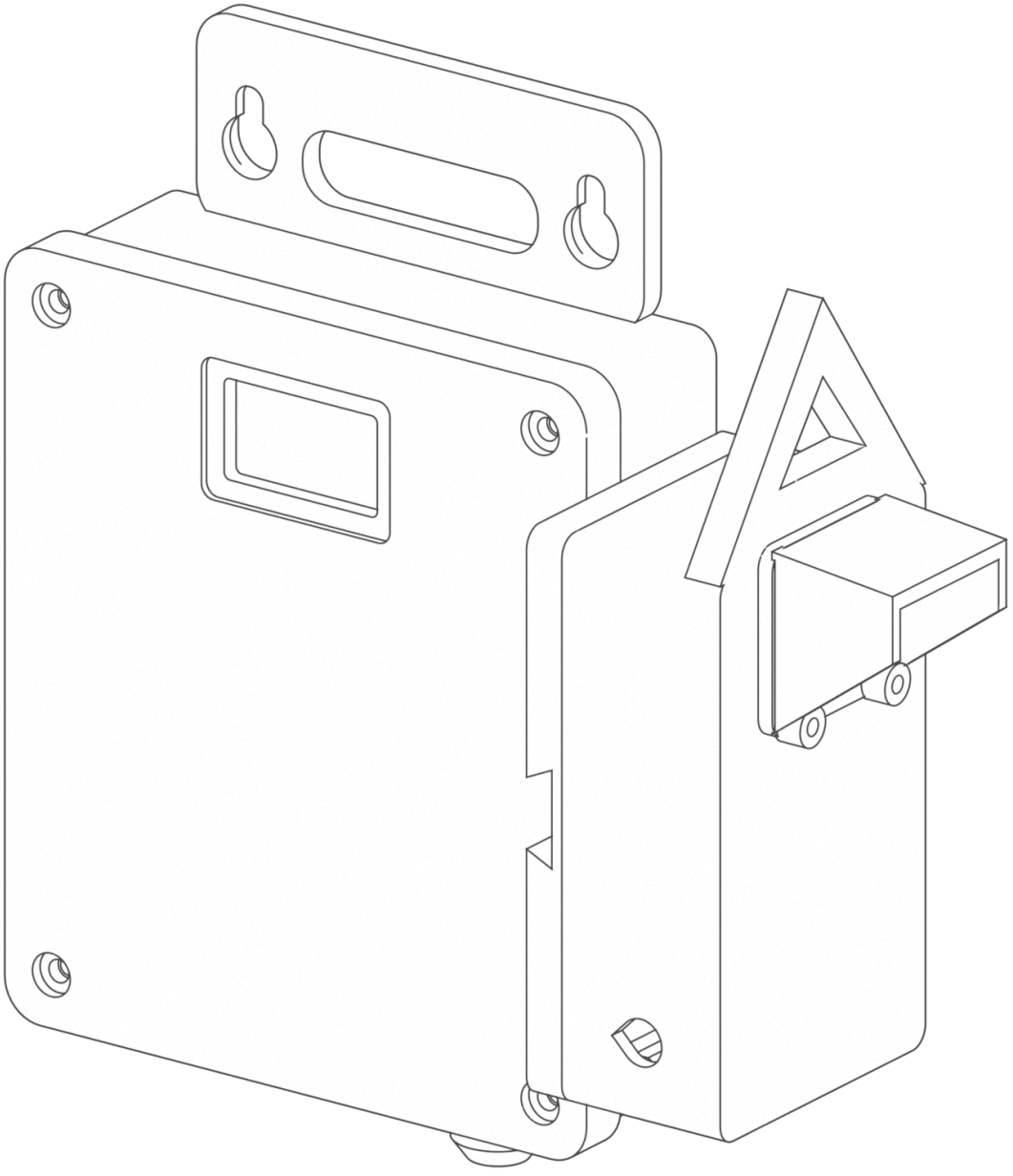
LÄRMBOX

A citizen noise sensor for Heilbronn

Overview catalogue · what it is, what is in the box, how it goes together

TEAM **Heilbronner Lilienwind**



1 person

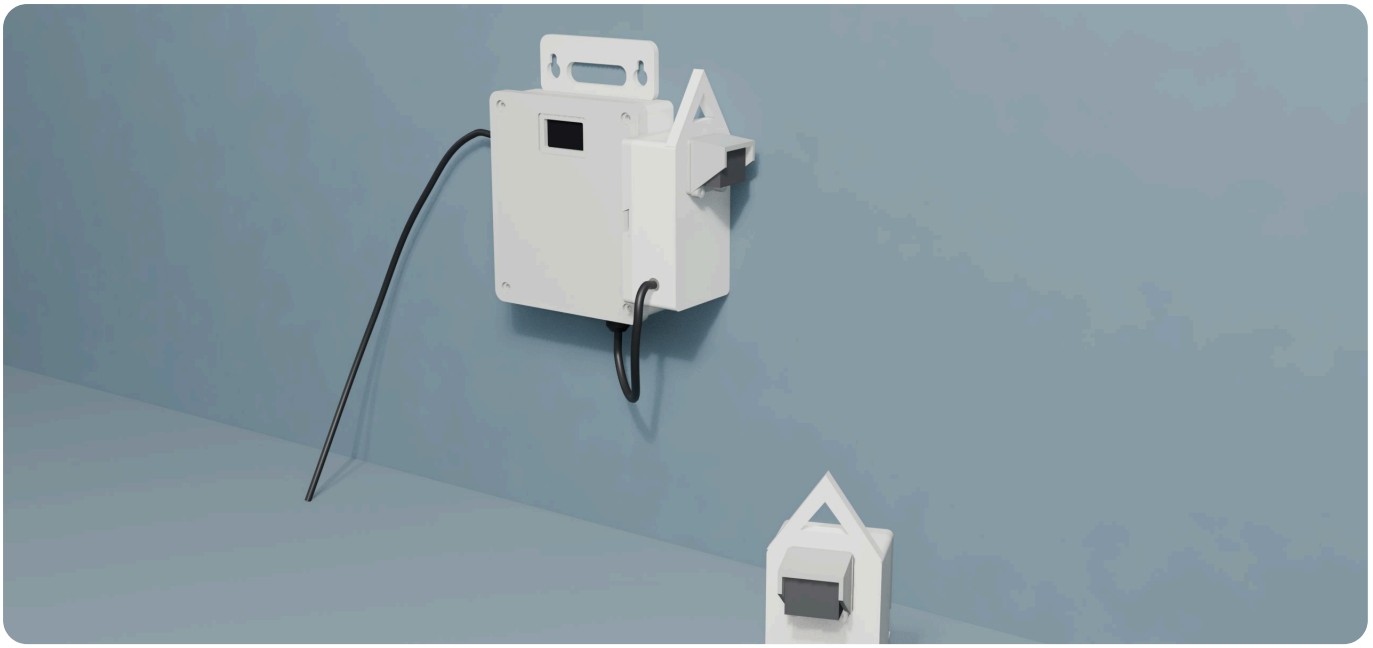
Phillips PH1

6 printed parts

Laptop

What it does

It hears the street, not the people.



1 Listen

The microphone pod measures the sound level **20 times per second**, A-weighted like the human ear.

2 Sum up

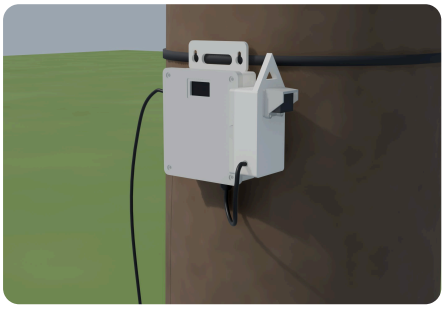
Every minute the box keeps three numbers: **LAeq** (energy average), **LAmax** (loudest moment), **LA90** (background).

3 Share

Over WiFi to openSenseMap, live on **laermlupe.de/dashboard**. Every second also goes to the SD card.

Only sound levels (dB) are stored. No audio is ever recorded.

One box, three places



Tree or pole · strap through the handle slot



Facade · two screws, keyholes

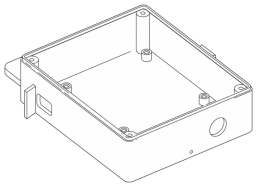


Walk · on the backpack chest strap

What's in the box

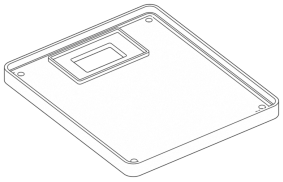
A–F printed · G–J electronics · K–L foam

A 1×



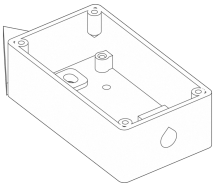
Main box

B 1×



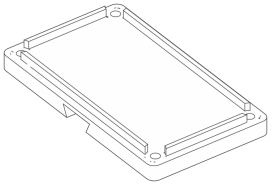
Lid with display window

C 1×



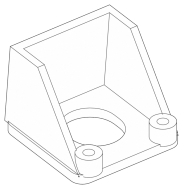
Microphone pod

D 1×



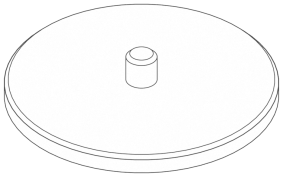
Pod back with slide slot

E 1×



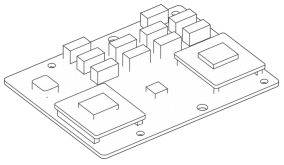
Rain hood

F 1×



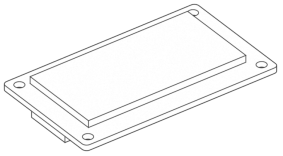
Foot (for the table)

G 1×



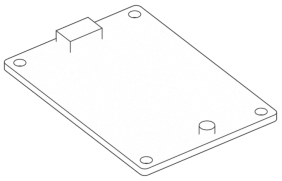
senseBox MCU + WiFi + SD

H 1×



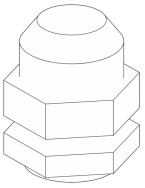
OLED display

I 1×



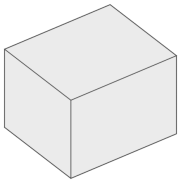
Sound level meter

J 1×



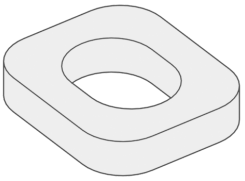
Cable gland PG7

K 1×



Foam block 22×19×16

L 1×



Foam ring 2–3 mm

Screws

drawn at about 1:1 · hold yours against the page



4×
M3×6



6×
M3×8



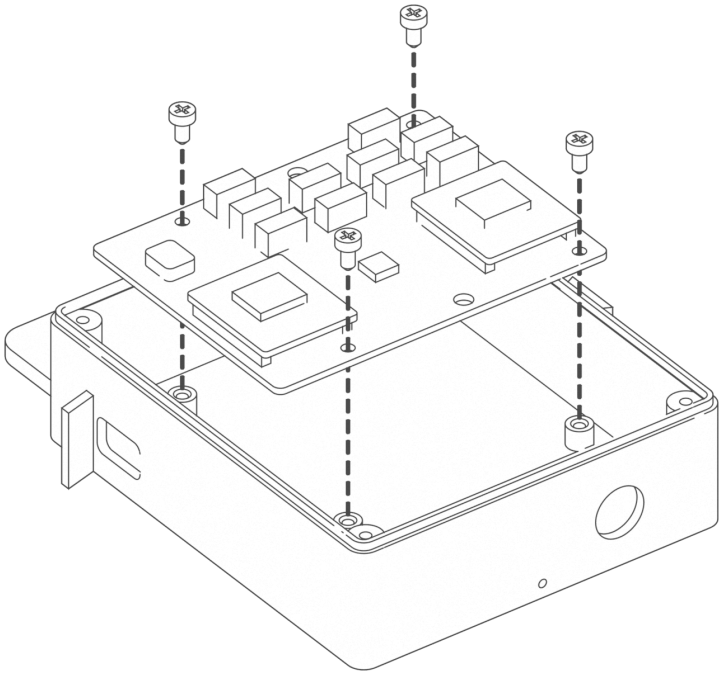
8×
M3×12



2×
Ø4 wall
+ plugs, optional

Also needed, not drawn: USB-C cable (carries data), the kit's JST cable for the display, the 3-pin → 4-pin cable for the sound meter (lengthened, soldered and heat-shrunk), double-sided tape. For a tree or a backpack: a 10 mm strap.

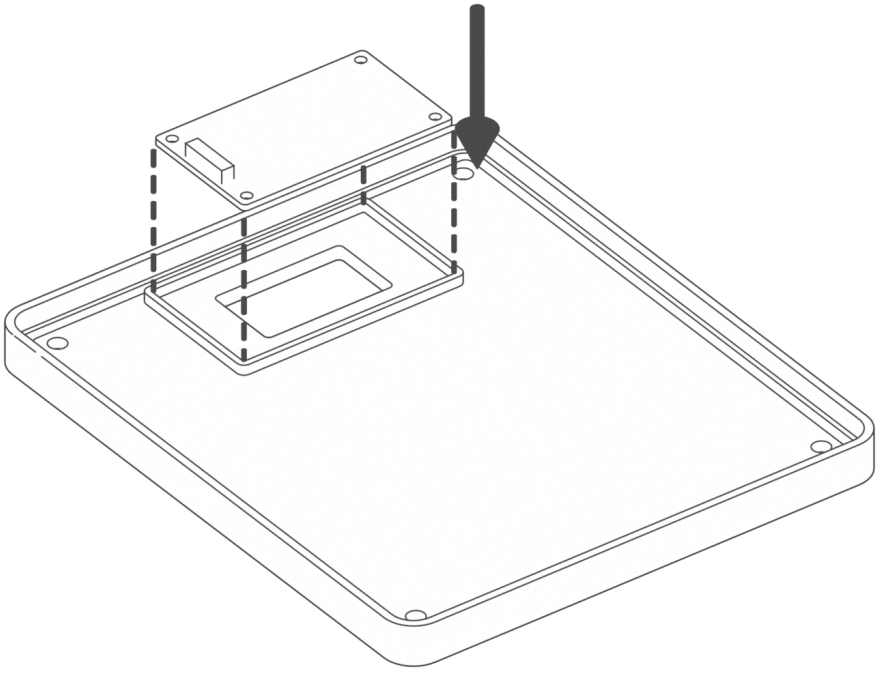
1 Screw the senseBox MCU onto the four posts in the main box (A). USB socket to the left.



4×
M3×6

WiFi-Bee and microSD-Bee are already on the board.

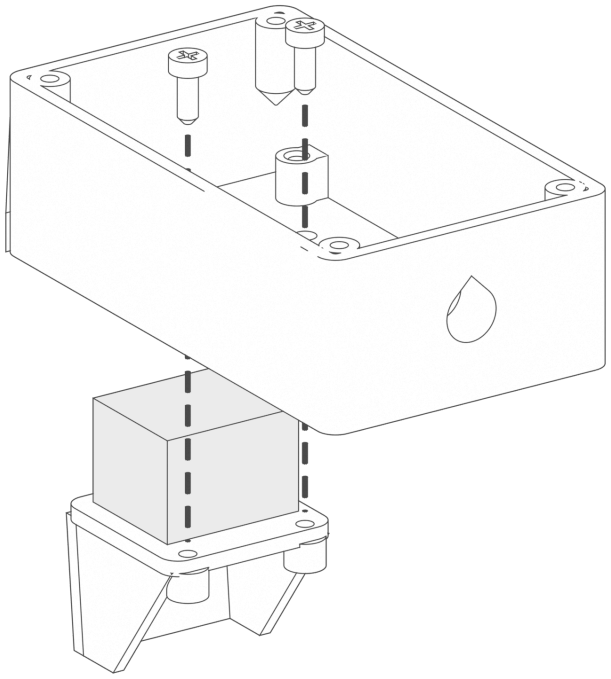
2 Stick the display (H) into the frame inside the lid (B), screen facing the window.



double-sided tape

Plug the JST cable into the display first.

3 Rain hood first: foam block (K) into the hood (E), then screw it on from inside the pod (C).

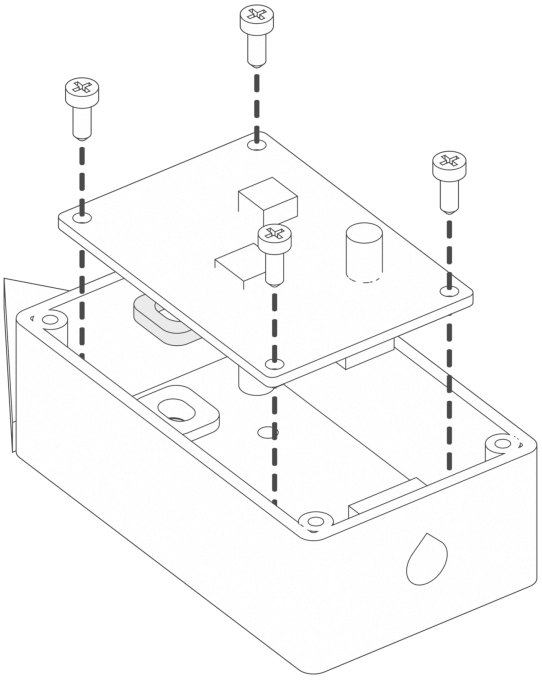


2×
M3×8

K

The hood's roof sits at the handle end: that is the top.

4 Foam ring (L) onto the round collar. Sound meter (I) on top, microphone facing down into the ring.

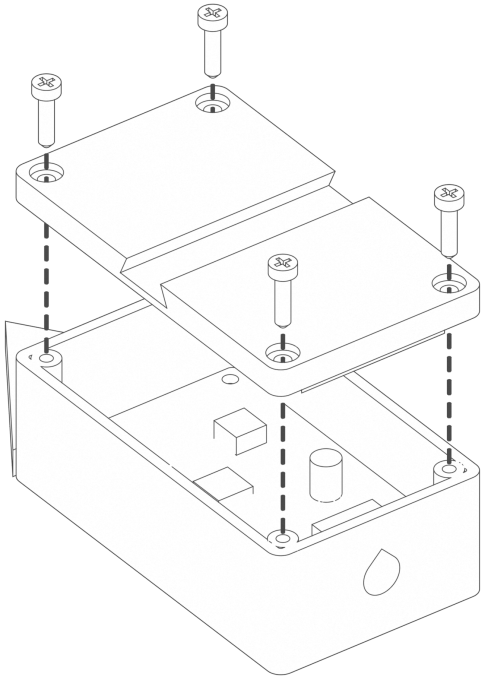


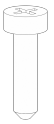
4×
M3×8

L

Plug the sound cable in before you close the pod.

5 Feed the sound cable out through the side hole, then close the pod with its back (D).

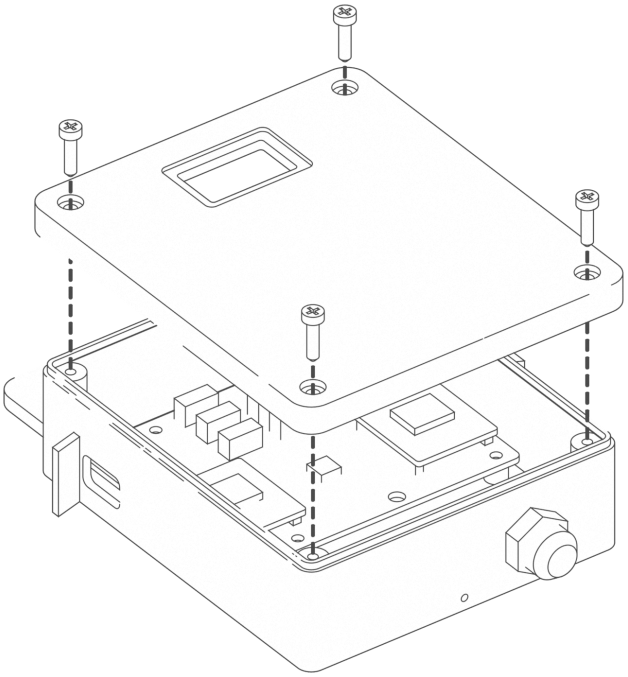




4×
M3×12

Seal the cable hole with a dab of silicone.

6 Sound cable through the gland (J) into port A, display cable into I2C. Then the lid.

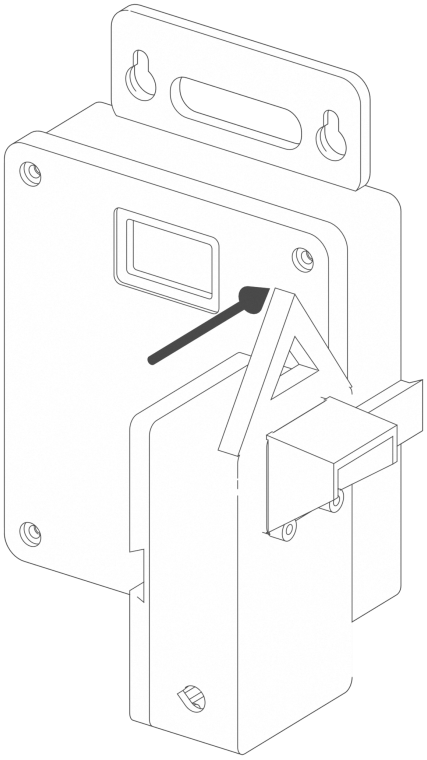




4×
M3×12

The USB-C plug goes in from outside, through the side window under the little roof.

Lid seal: a bead of silicone or a TPU ring in the groove keeps rain out.



7

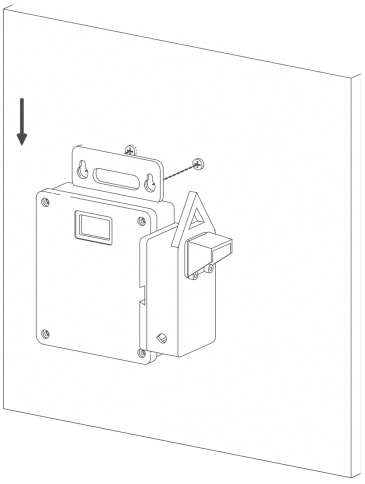
Slide the pod onto the rail on the side of the box until it stops.

The rail runs from front to back, so the pod slides on from the lid side. Want the microphone somewhere else? Pull the pod off and put it on its foot (F).

No tools. A snug fit holds it; a drop of glue makes it permanent.

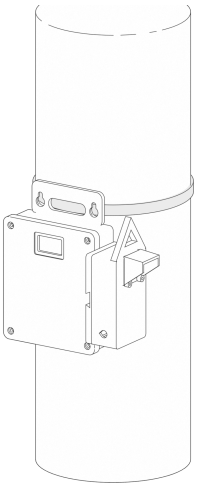
8

Pick a place. Microphone towards the street.



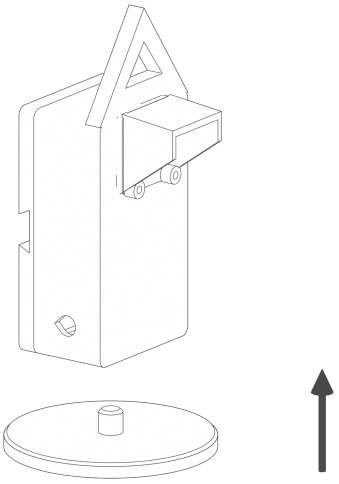
8a · Wall

2 screws Ø4, 58 mm apart. Push on, slide down.



8b · Tree or pole

10 mm strap through the slot. No screws in trees.



8c · Table

Pod on its foot, turns 360°.

Not behind glass, not inside a bag. On a facade the wall reflects sound and adds about 3 dB: note where the box hangs. Ask before you use someone's wall or window.

Switch on. See it live.

Plug in USB-C. The display shows the level right away; after a minute the first value is online.

LIVE DASHBOARD

laermlupe.de/dashboard

Every LärmBox on a map, the loudest minute right now, and a curve of the last hours next to the WHO reference lines.

LAeq

The energy average of the minute. The number most noise rules use.

LAmax

The loudest moment: a truck, a horn, a motorbike.

LA90

The background: what is left when nothing passes by.

Open data. Values go to openSenseMap; the campaign dataset is published under CC BY 4.0.

Open source. Firmware, pipeline, dashboard and these 3D parts: `github.com/itsmehmettaskin/ greencity-hackathon-laermlupe`

Only sound levels (dB) are stored. No audio is ever recorded. No private addresses on the map.

Team Heilbronner Lilienwind · Özgün Mert Şahin, Mehmet Taşkın, Eren Mutlu, Cahit Gerçekel

LärmLupe
by Heilbronner Lilienwind